

Pavithra Priyadarshini Selvakumar

Phone: +1 919-638-7355

Email: pselvak@okstate.edu

LinkedIn: <https://www.linkedin.com/in/pavithra-priyadarshini>

Education

Postdoctoral Research Scientist

Columbia University

September 2025 - Present
New York City, New York

Doctor of Philosophy in Environmental Science

Oklahoma State University

Dissertation: Institutional sustainability in U.S. Land-Grant Universities:

A Comparative Study of Oklahoma State University (OSU) And

Colorado State University (CSU)

June 2025
Stillwater, Oklahoma
CGPA (4.0/4.0)

Master of Science in Environmental Assessment and Management

University of East Anglia

September 2018
Norwich, United Kingdom

Bachelor of Technology in Information Technology

Anna University

May 2015
Chennai, India

Research Experience

Postdoctoral Research Scientist

Columbia Climate School

September 2025 – Present
New York City, New York

- Researching equitable coastal adaptation, AI, and climate justice under the Resilient Coastal Communities Project (RCCP).
- Collaborating with the New York City Environmental Justice Alliance (NYC-EJA), the Center for Sustainable Urban Development (CSUD), and the Lamont-Doherty Earth Observatory (LDEO) to strengthen partnerships between scientists, policymakers, and frontline communities for inclusive and data-driven flood protection planning.
- Working on a systematic review paper on AI and Climate Justice, analyzing how artificial intelligence can support equitable climate adaptation and mitigate algorithmic bias in decision-making.
- Working on a “Transactional to Relational Engagement” article focusing on fair and accountable coastal adaptation.
- Mentoring an undergraduate student on qualitative data analysis, and community engagement.

Dissertation on Campus Sustainability

Oklahoma State University

August 2021-June 2025
Stillwater, Oklahoma

- Conducted semi-structured qualitative interviews with diverse stakeholders from OSU and CSU, including staffs, faculties, students, and community members, to collect data on campus sustainability practices.
- Performed iterative coding based on drivers, barriers, leadership, and other critical factors influencing sustainable practices in higher educational institutions.
- Conducted Group Model Building (GMB) sessions with stakeholders to gain insights into their perceptions, challenges, and suggestions regarding sustainability efforts on campus.

- Analyzed data using software such as Nvivo to identify patterns, trends, and areas for improvement in campus sustainability efforts.
- Developed recommendations for enhancing sustainability practices on campus based on research findings.

Graduate Research Assistant - NSF EPSCoR

Social Dynamics research team

August 2021-May 2025

Oklahoma

- Worked with Opinion Leader Advisory Network (OLAN) and Extended Peer Scientific Advisory Network (EPSAN) to provide socially sustainable solutions to ‘wicked’ environmental problems of Oklahoma.
- Conducted focus group discussions and GMB sessions to tackle Oklahoma’s complex environmental challenges, engaging stakeholders in developing causal loop diagrams and mental models to identify sustainable solutions.
- Applied Advocacy Coalition Framework (ACF) and the Narrative Policy Framework (NPF) for analyzing policy processes and environmental governance.
- Assisted in the preparation of research reports, presentations, and journal publications.
- Participated in meetings, workshops, and conferences (research retreats and academies) to share research findings and insights.
- Facilitated the student-peer networking breakout session in the Fall Retreat 2021.

Research Experience (continued)

Group Model Building – Tribal Nation Community Engagement Project

Facilitator & Chart Writer

Fall 2022

Pawnee Nation, Oklahoma

- Facilitated GMB sessions among the key stakeholders of Pawnee Nation Tribal community.
- Collaborated in a dynamic team setting to develop strategies for a ‘Commercial Kitchen’.
- Identified four key drivers for agricultural development: extension, general knowledge, funding, and a business plan, highlighting the overlaps and stressing the importance of each theme.
- Contributed to discussions on sustainability and community support through various initiatives, including regenerative grazing, small-scale vegetable production, and educational outreach.
- Used mental models and causal loop diagrams to understand the themes discussed during the GMB session.
- Identified the drivers and barriers that could promote sustainability in the community.

Environmental Humanities Research Fellow

Center for the Humanities, Oklahoma State University

August 2023-2024

Stillwater, Oklahoma

- Developed "Moving Orange in a Green Way", a project dedicated to promoting environmentally sustainable practices within the Stillwater community.
- Co-developed an upcoming course titled "Environmental Thinking and Philosophy," aimed at disseminating regional environmental ideologies and policy frameworks.
- Collaborated on the creation of a rideshare application to improve sustainable transit solutions for the campus community.
- Contributed to "Building the Bigger Table", an initiative designed to encourage dialogue on food security and strengthen community bonds via collective storytelling and shared culinary experiences.

Summer School in Computational Social Sciences

University of Rochester

Summer 2024

Rochester, New York

- Participated in lectures and tutorials on image and text as data, social media APIs, data visualization, web scraping, mass collaboration, network analysis, machine learning, and deep learning.

- Developed a project that conducts geospatial analysis on campuses' green spaces, energy efficiency, water conservation, waste management, and transportation.
- Utilized web scraping techniques and the YOLO (You Only Look Once) object detection model for data collection and analysis.
- Created an independent ranking system based on geospatial analysis and compared it with AASHE STARS ratings to evaluate correlations and identify discrepancies.
- Provided data-driven recommendations to improve campus sustainability and achieve climate action goals.
- Wrote a paper titled "Geospatial Analysis of Campus Sustainability of Universities in the USA."

Master's Dissertation Research

University of East Anglia

December 2018
Norwich, United Kingdom

- Investigated the role of Corporate Social Responsibility (CSR) in Hospitals.
- Performed extensive literature review on the research background and related problems.
- Conducted a qualitative case study on Serco (a public service firm) in Norfolk and Norwich University Hospital.
- Integrated staff questionnaires and analyzed company data and documents.
- Analyzed data using Excel and SPSS.
- Identified various drivers and barriers for the company to engage in environmental projects.

Paper Publications

1. Sibley, M., Peach, K., León-Corwin, M., **Selvakumar, P. P.**, Diodosio, K., Fox, A., ... & Olofsson, K. (2024). Exploring risk-scapes in Oklahoma: Institutional Trust, Environmental Justice, Climate Change, and Infrastructure. *Safer Communities*.
2. Olofsson, K. L., **Selvakumar, P. P.**, Peach, K., Leon-Corwin, M., Stormer, S. A., Gupta, K., ... & Sibley, M. (2023). Effective Stakeholder Engagement in Environmental Problem-Solving through Group Model Building: An Oklahoma Case Study. *Environmental Challenges*, 100755.
3. **Selvakumar, P. P. (2020)**. Investigation of Corporate Social Responsibility in Multi-National Companies to Promote Environmental Sustainability. *OIDA International Journal of Sustainable Development*, 13(05), 11-20.
4. 'What Does Learning Look Like? A Methodological Advancement to Measure Learning in the Advocacy Coalition Framework (ACF)' – Submitted to Policy Studies Journal
5. 'Characterless: Developing Macro Narratives within the Narrative Policy Framework.' – Submitted to Policy Studies Journal.
6. 'Who Learns How: An ACF Study Exploring Individual Attributes of Climate Change Learners' – Work in Progress
7. 'Home Sweet 'Homa: Place Attachment, Political Participation, and the Environment in Oklahoma' – Work in Progress

Conference Presentations

1. **Midwest Political Science Association 2025** Chicago, Illinois
'Who Learns How: An ACF Study Exploring Individual Attributes of Climate Change Learners'
2. **National Sustainability Society 2024** Seattle, Washington
'Sustainability Performance and Leadership in Higher Educational Institutions: A Comparative Examination between Oklahoma State University (OSU) and Colorado State University (CSU)'
3. **XVIII World Congress of Comparative Education Societies** Cornell University, New York
'Gamifying Campus Sustainability through a Unified Digital Ecosystem'
4. **Oklahoma EPSCoR S30K Annual State Conference 2024** Oklahoma City, Oklahoma
'From Outages to Outlook: Power Outages and Trust in Institutions'
5. **Southern Political Science Association 2023** St. Pete's Beach, Florida
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‘Changing Your Mind: Testing the Flexibility of Individual Policy Beliefs’

6. **Oklahoma Clean Lakes and Watersheds Association 2023** Stillwater, Oklahoma
‘Collaboration for socially sustainable solutions for water issues in Oklahoma.’
7. **Oklahoma EPSCoR S3OK Annual State Conference 2022** Tulsa, Oklahoma
‘Collaboration for socially sustainable solutions for water, carbon and infrastructure resilience in Oklahoma.’
8. **International Conference on Sustainable Development 2019** Harvard University, Boston
‘Investigation of Corporate Social Responsibility in Multinational companies to promote Environmental Sustainability’
9. **OK NSF EPSCoR Research Retreat** Midwest City, Oklahoma
‘Cross Stakeholder Assessment of Causes and Consequences of Environmental Problems in Oklahoma’
10. **The Asian Conference on Sustainability, Energy & the Environment 2019** Virtual Presentation
‘Role of Corporate Social Responsibility in Hospitals to Improve Environmental Sustainability’

Grants Received (Total \$17,480)

- Awarded the \$8000 Robberson Summer Dissertation Fellowship 2024 for dissertation research support.
- Received \$1,980 funding from the Fae Rawdon Norris Grant Committee for the "Foods of Home – Building a Bigger Table" event featuring guest speaker Dr. Gisela Heffes for the Fall 2024-Spring 2025 period.
- Granted the prestigious 2023-24 Humanities Research Group Fellowship, providing funding of \$3000 to pursue research in the humanities field.
- Received the 'Grand Prize Scientific Poster Presentation' and a \$2000 cash award for delivering a compelling presentation on 'Cross Stakeholder Assessment of Causes and Consequences of Environmental Problems in Oklahoma' at the EPSCoR S3OK Research Retreat.
- Secured a \$2000 travel grant from the Department of Environmental Science to support attendance at conferences and professional development activities.
- Received conference funding of \$300 from the Graduate and Professional Student Government Association to present research findings at a professional conference.
- Received \$200 for second place in Oklahoma State University Abstract Writing Competition.

Leadership Experience

Student Executive Board Member – NSF EPSCoR Spring 2022 – May 2025

- Orchestrated the Student Professional Development Session at the Oklahoma NSF EPSCoR 2024 Annual Conference, facilitating career advancement and professional skills enhancement for attendees.
- Organized a seminar series to foster academic and professional development among peers.
- Organized an OSU/OU Football Game Watch Meet-Up, facilitating networking in an informal setting.
- Contributed to the formulation of the board’s mission statement and future strategic initiatives.
- Coordinated the student poster session for the EPSCoR annual meeting in April 2023.

Vice President – Society for Environmental Scientists Fall 2023 - Present

- Elected as Vice President to steer the club's initiatives, driving engagement and promoting environmental awareness within the university community.
- Led the planning and execution of educational events, workshops, and community service projects.
- Facilitated collaboration between students and faculties to enrich the club's activities and impact.

Other Relevant Work Experience

- Conducted regular monitoring of water quality parameters in the city of Stillwater.
- Assessed potential contaminants and ensured the safety and compliance of the water supply.
- Implemented a comprehensive water sampling program in coordination with local authorities.
- Collected representative samples from various sources to assess key water quality indicators.
- Identified any potential issues or deviations from established standards.

Project Coordinator – ‘Budget Bites’ Project

Enactus UEA

Sep 2017 - Sep 2018
Norwich, United Kingdom

- Conceptualized and implemented ‘Budget Bites’ project, addressing food waste and food poverty in Norwich.
- Sourced surplus food by establishing partnership with the Norwich Food Hub.
- Contributed to the upcycling of collected food items, offering them to the community on a 'pay as you feel' basis to ensure accessibility to affordable food.
- Successfully launched two pop-up stalls, leading to the development of the Budget Bites Café concept aimed at providing affordable meals and work experience opportunities.
- Supported the project's alignment with the United Nations Sustainable Development Goals, specifically targeting responsible consumption and production, zero hunger, and decent work and economic growth.
- Presented the ideas of the project ‘Budget Bites’, an idea of sourcing food surplus and upcycling them for consumption, in Enactus UK National Expo held in London.
- Fostered community relations by organizing events such as pop-up stalls, training, and workshops. uniting instruction, and variety to garner interest and summon greater participation.

International Students Ambassador

University of East Anglia

Jan 2018 - Nov 2018
Norwich, United Kingdom

- Supported the International Office in administrative activities and responsibilities.
- Assisted International Students with prompt and clear communications about campus developments.
- Networked with prospective students through email accounts and social media platforms.
- Promoted diversity, equity and inclusion by conducting deeper engagement with the future students.
- Arranged campus tours for the incoming students.

Engineer (Sustainable Development Team)

Tata Consultancy Services

June 2015 – Aug 2017
Chennai, India

- Developed user interface to track energy consumption in mobile applications.
- Contributed to rein in company’s carbon and resource footprint using designing and technical skills.
- Used technologies including React.js, JavaScript, Python and CSS.
- Assessed the organization’s sustainability performance using various statistical skills.
- Identified the ‘limited access to energy and water in the campus’ as the organization’s key sustainability issues that led to reduced productivity at the workplace.
- Disseminated awareness about responsible consumption of water and electricity at the workplace through floor walks.
- Collaborated with the company’s internal groups and developed sustainability plans through focus group discussions.

Workshops Attended and Community Contributions

- Peer reviewed 6 articles for the International Journal of Sustainability in Higher Education, recognized and indexed by the Web of Science (ResearcherID: IWU-9586-2023).
- Provided constructive feedback to undergraduate researchers during the third annual OSU Undergraduate Research Symposium poster session, helping to enhance their presentation and research skills.
- Spearheaded sustainability initiatives as Net Impact Graduate Chair, OSU Chapter, fostering a more inclusive and equitable environment at OSU.
- Attended the Fall 2024 Graduate Teaching Assistants conference and gained knowledge in effective instructional techniques, contributing to the development of my teaching portfolio.
- Attended conference workshops on Collaborative & International Grants, Leadership in 3D and Improving your Publication Success conducted by KiND Institute.
- Completed a workshop on *Closing the loop: why do we need post occupancy evaluation for buildings* at the Appalachian State University, Boone, North Carolina (September 2018).
- Participated in the *Eastern Arc Start-up Weekender 2019* conducted by Eastern Academic Research Consortium that created an opportunity to drive growth and innovation in the project ideas through various activities and academic expertise.
- Participated in the Barclays Accelerator Programme hosted by key decision makers at Barclays to provide guidance for innovative solutions in technological world.
- Headed the project team that was selected in 'Manthan National Competition' held in 2013. Presented our ideas on 'Can Crusher', a proposal for a mechanism that could reduce crushing cost incurred for cans and bottles.

Awards/Certificates

- Awarded *International Student Academic Excellence Scholarship* for outstanding academic performance and extracurricular involvement at the University of East Anglia.
- *Top Cultural Ambassador Award 2018* of University of East Anglia for demonstrating cultural sensitivity and appreciation through work.
- *Top Performer Award 2018* of Enactus UEA for extraordinary contribution for the project 'Budget Bites'
- *Special Initiative Award – 2016* of Tata Consultancy Services for spreading awareness through 'floor walks' on responsible consumption of water and electricity at the workplace among the employees.
- *Certification in Human Resource and Business Analytics (2015)* of Loyola Institute of Business Administration (LIBA), for completing courses in statistical computing, data analytics and scientific research used by researchers to retrieve, clean, analyze, visualize and present data.

Skills and Interests

Computer Skills: Proficient in Vensim, Stata, ArcGIS, IBM SPSS, R programming, React.js, Python, JavaScript, CSS, Latex, and Microsoft Office.

Language Skills: Native proficiency in Tamil, professional proficiency in English, basic proficiency in Hindi, Telugu, and Malayalam.